

EXPT NO:6	INTEGRATE DATA FROM VARIOUS SOURCES FOR COMPREHENSIVE ANALYSIS USING TABLEAU AND POWERBI
DATE: 24.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)**PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)****1. What is data integration in the context of Business Intelligence (BI)?**

Data integration is the process of combining data from multiple sources into a single, unified view for analysis and reporting. In Business Intelligence, data may come from databases, Excel files, CSV files, cloud platforms, or web services. Integrating these sources ensures that all relevant information is available in one place, making analysis more accurate and meaningful. It also helps eliminate duplicate data, reduces inconsistencies, and improves decision-making. A well-integrated dataset provides a complete picture of business performance and enables organizations to generate reliable insights.

2. Explain the difference between a join and a data blend in Tableau.

A join combines two or more tables at the data source level based on a common field, resulting in a single table before analysis begins. It is suitable when the tables come from the same or compatible data sources. In contrast, data blending combines data from different data sources after the data has been loaded into Tableau. It links the data during visualization rather than physically merging the tables. Joins generally provide better performance, while data blending is useful when data cannot be directly joined because it originates from different sources.

3. What are relationships and cardinality (e.g., one-to-many, many-to-many) in Power BI's data model?

Relationships in Power BI define how tables are connected using common fields so that data can be analyzed across multiple tables. Cardinality describes the type of relationship between those tables.

- One-to-One (1:1): One record in each table matches exactly one record in the other table.
- One-to-Many (1:*) : One record in the first table can be related to multiple records in the second table. This is the most commonly used relationship.
- Many-to-Many (:): Multiple records in one table can relate to multiple records in another table.

Choosing the correct relationship and cardinality ensures accurate filtering, aggregation, and reporting.

4. Why is data quality (e.g., consistent naming conventions, data types) crucial when integrating data from different sources?

Good data quality is essential because inaccurate or inconsistent data can lead to incorrect analysis and misleading business decisions. When integrating data from different sources, differences in column names, data types, or formats can create errors during joins and relationships. Maintaining consistent naming conventions and proper data types ensures that data is combined correctly and can be analyzed without issues. High-quality data also improves the reliability of reports, reduces the need for manual corrections, and increases confidence in the insights generated from the dashboard.

5. Name two common challenges encountered during data integration.

Some common challenges faced during data integration include:

- Inconsistent data formats and data types: Different sources may store the same information in different formats, making integration difficult.
- Missing or duplicate data: Incomplete records or duplicate entries can reduce the accuracy and reliability of the final analysis.

These challenges are usually addressed through proper data cleaning, transformation, and validation before analysis.

IN-LAB EXERCISE in tableau

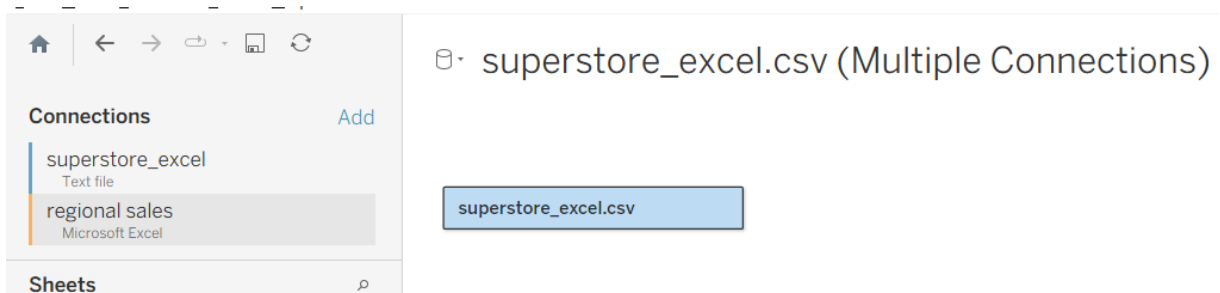
OBJECTIVE:

To connect to, combine, and model data from multiple disparate sources in Tableau and Power BI to create a unified, comprehensive dashboard for holistic analysis.

PROCEDURE:

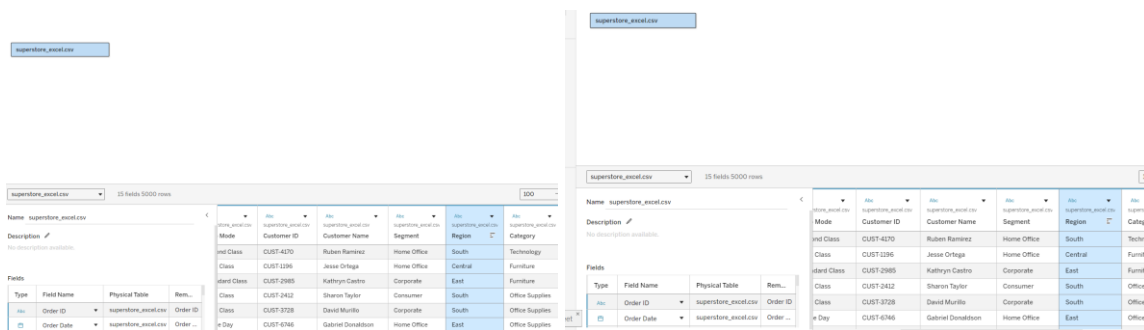
1. Data Collection:

- Import data from at least two different sources (e.g., a CSV file and an Excel file).
- **Dataset 1: Sales Data (CSV file) with fields like Order ID, Order Date, Region, Sales Amount.**
- **Dataset 2: Regional Sales Targets (Excel file) with fields like Region, Sales Target, Target Year.**



2. Data Preparation and Transformation:

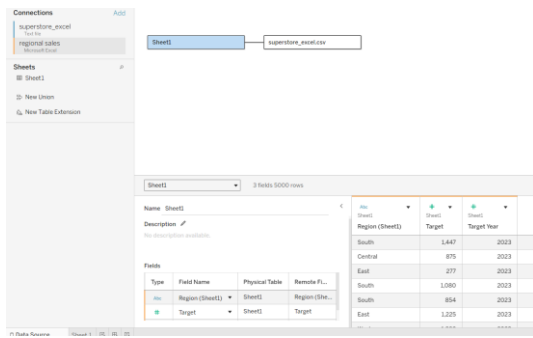
- Clean both datasets: Check for inconsistencies in region names (e.g., "N. America" vs. "North America").
- Ensure data types are correct (e.g., Sales Amount is numeric, Order Date is a date).



- Handle any missing values as appropriate.

3. Data Integration and Modeling:

- **Tableau: Use the Data Source pane to establish a relationship or join between the Sales Data and Regional Targets tables using the common Region field.**

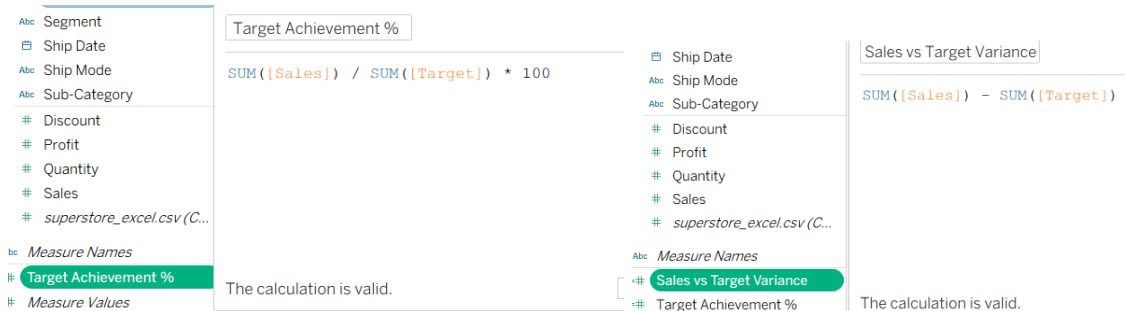


- **Power BI: Use the Model view to create a relationship between the two tables, connecting them on the Region column. Define the cardinality (e.g., many-to-one).**

4. Create Calculated Fields for Comparison:

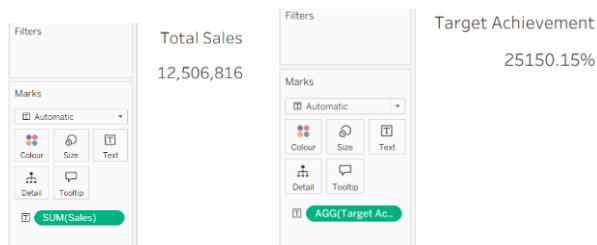
- **Develop new measures that leverage both datasets.**

- **Examples: Target Achievement % = (SUM(Sales) / SUM(Sales Target)) * 100**
- **Sales vs. Target Variance = SUM(Sales) - SUM(Sales Target)**

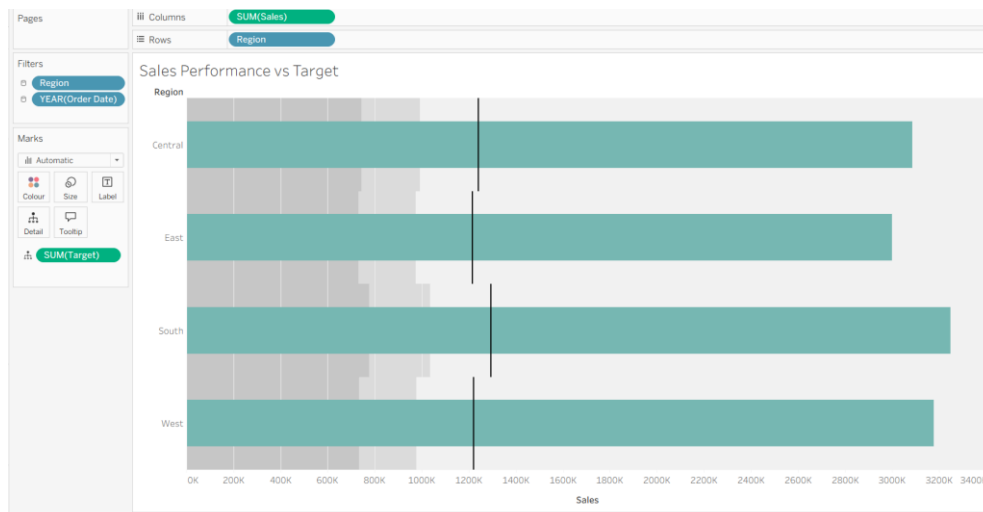


5. Design a Comprehensive Dashboard:

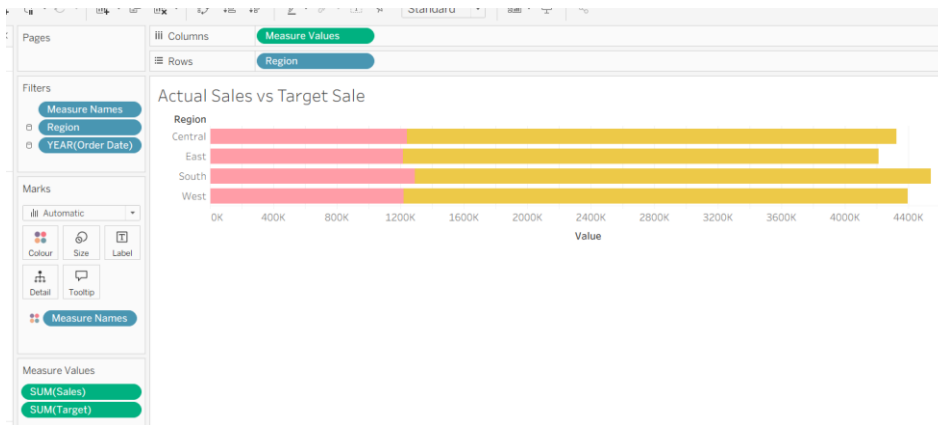
- **Create visuals that combine data from both sources to provide a single view of performance.**
- **Add KPI cards for Total Sales and Overall Target Achievement %.**



- **Use a gauge chart or bullet chart to show sales performance against targets for each region.**

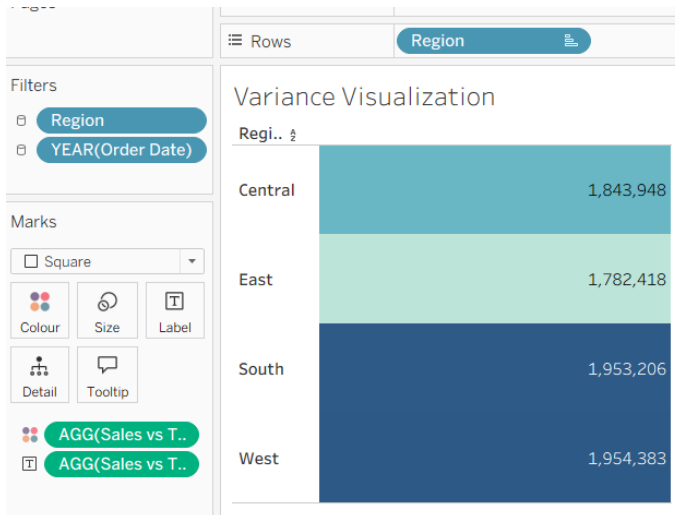


- **Create a combined bar chart showing Actual Sales vs. Target Sales by Region.**

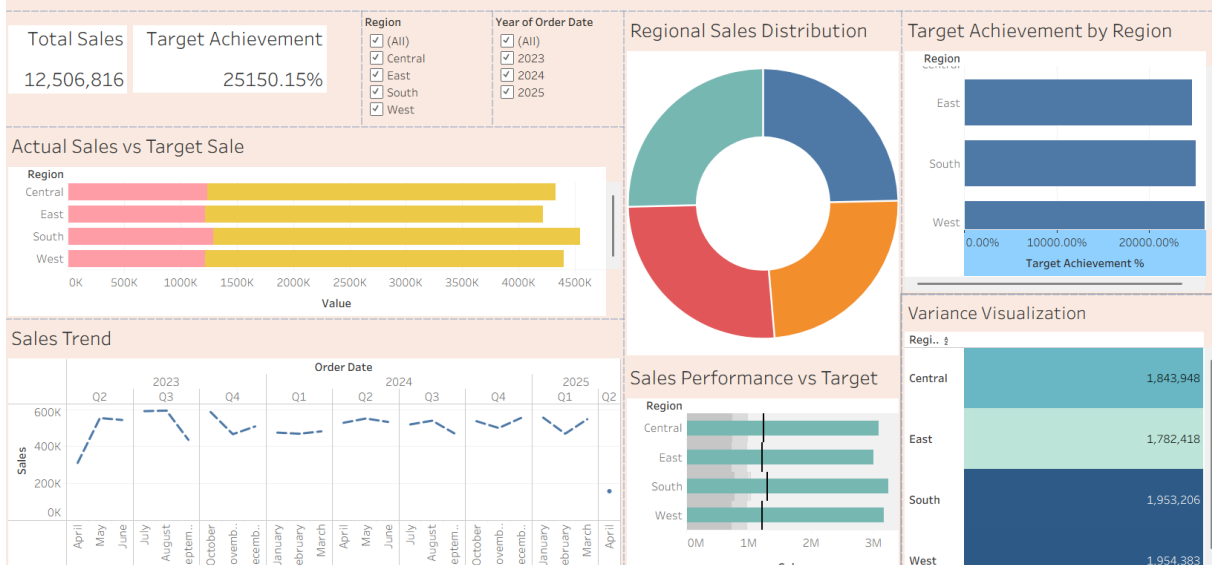


- Use a map to visualize the Sales vs. Target Variance, highlighting underperforming or overperforming regions

The provided dataset contains business regions (East, West, Central, South) instead of geographic locations such as States or Cities. Since Tableau requires geographic fields for mapping, a highlight table was used to visualize Sales vs Target Variance.



SALES TARGET PERFORMANCE DASHBOARD



6. Validate Data Integrity:

- **Verify that the joined/related data is accurate by cross-checking values.**
- **Ensure filters on the dashboard correctly update visuals that use data from both sources.**

7. Publishing and Sharing:

- **Tableau: Publish the workbook to Tableau Public or Tableau Server.**
- **Power BI: Publish the report to Power BI Service and share the link.**

Experiment Datasets:

Dataset 1: Retail Sales Data (CSV)

Fields Used:

- **Order ID – Unique identifier for each order**
- **Order Date – Date of the sales transaction**
- **Region – Sales region**
- **Sales – Sales revenue from the order**

Dataset 2: Regional Sales Targets (Excel)

Fields Used:

- **Region – Sales region (used as the key for joining/relationship)**
- **Sales Target – The sales goal for the region**
- **Target Year – The year for which the target is set**

IN-LAB EXERCISE in powerBI:

OBJECTIVE:

To connect to, combine, and model data from multiple disparate sources in Tableau and Power BI to create a unified, comprehensive dashboard for holistic analysis.

PROCEDURE:

8. Data Collection:

- **Import data from at least two different sources (e.g., a CSV file and an Excel file).**
- **Dataset 1: Sales Data (CSV file) with fields like Order ID, Order Date, Region, Sales Amount.**
- **Dataset 2: Regional Sales Targets (Excel file) with fields like Region, Sales Target, Target Year.**

Region	Target	target year
South	1447	2023
Central	875	2023
East	277	2023
South	1080	2023
South	854	2023
East	1225	2023
West	1382	2023
East	287	2023
South	1147	2023
South	930	2023
Central	512	2023
South	1729	2023
Central	173	2023
East	550	2023
South	513	2023
South	1427	2023
Central	1037	2023
South	562	2023
East	1021	2023
West	618	2023
Central	1676	2023
South	848	2023
East	1453	2023
East	-78	2023

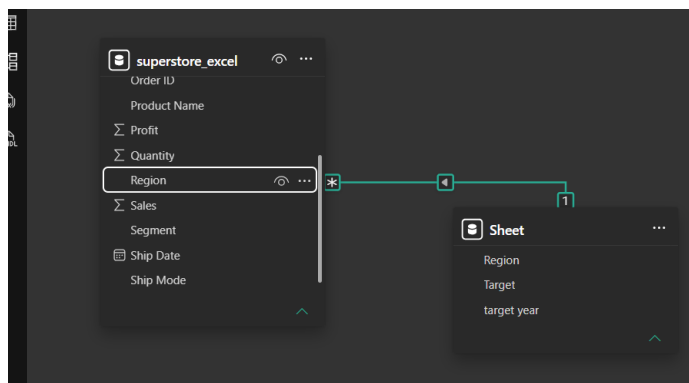
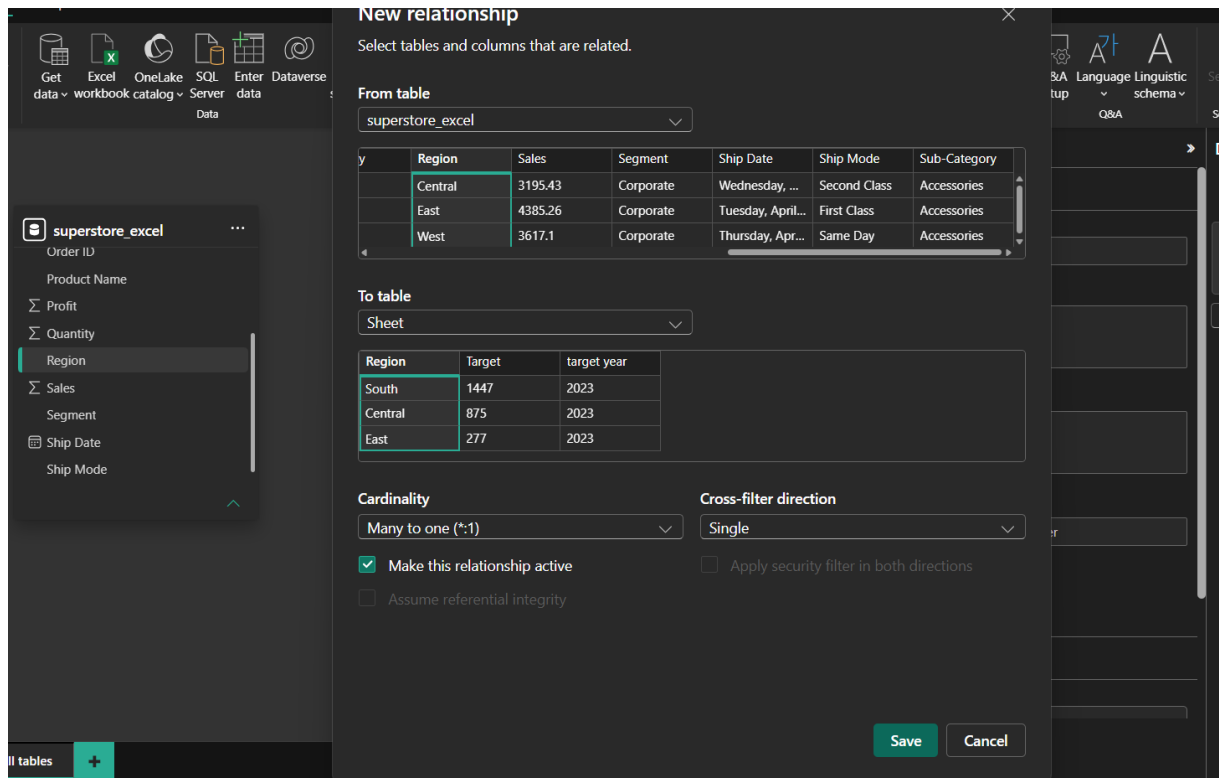
Customer ID	Customer Name	Segment	Region	Category	Sub-Category
1	Roben Ramirez	Home Office	South	Technology	Copiers
2	Jesse Ortega	Home Office	Central	Furniture	Furnishings
3	Kathryn Castro	Corporate	East	Furniture	Chairs
4	Sharon Taylor	Consumer	South	Office Supplies	Labels
5	David Murillo	Corporate	South	Office Supplies	Binders
6	Gabriel Donaldson	Home Office	East	Office Supplies	Storage
7	Colleen Rose	Corporate	West	Office Supplies	Storage
8	Grant Garcia	Home Office	East	Office Supplies	Storage

9. Data Preparation and Transformation:

- **Clean both datasets:** Check for inconsistencies in region names (e.g., "N. America" vs. "North America").
- **Ensure data types are correct** (e.g., Sales Amount is numeric, Order Date is a date).
- **Handle any missing values as appropriate.**

10. Data Integration and Modeling:

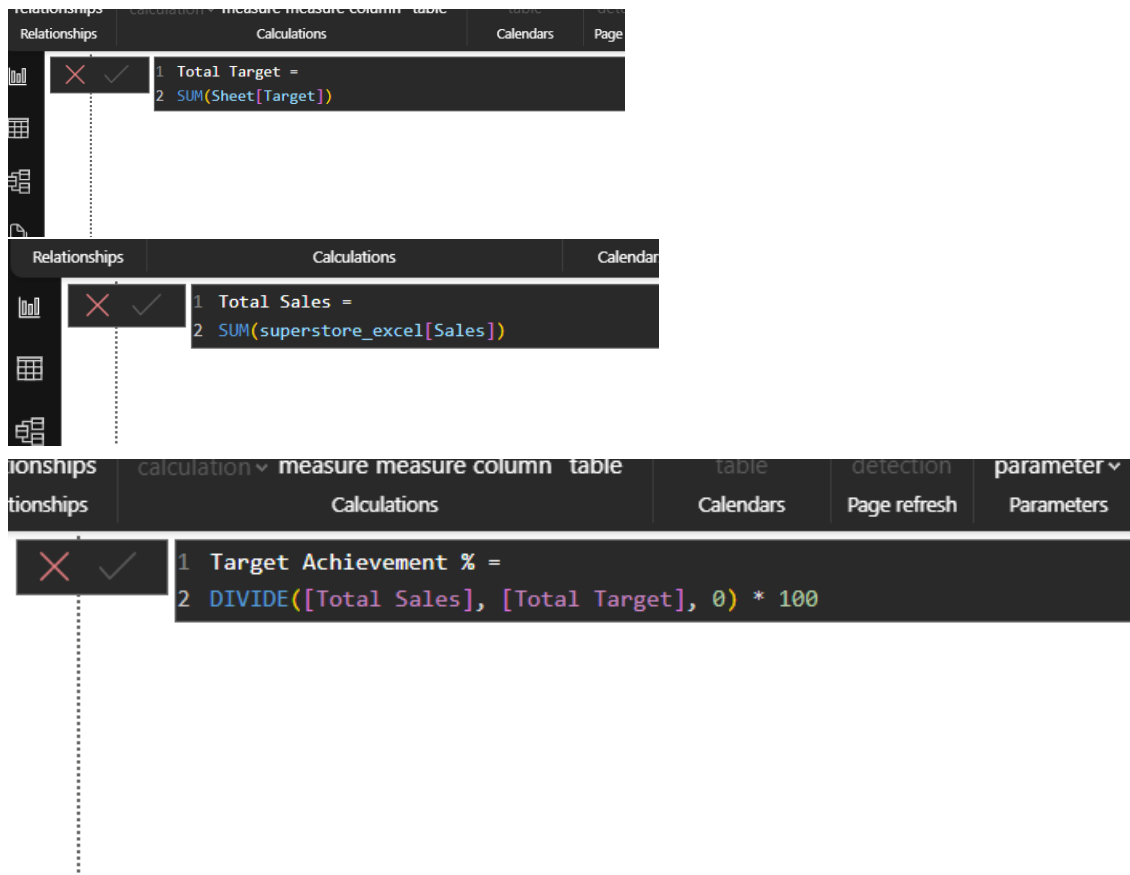
- **Power BI:** Use the Model view to create a relationship between the two tables, connecting them on the Region column. Define the cardinality (e.g., many-to-one).



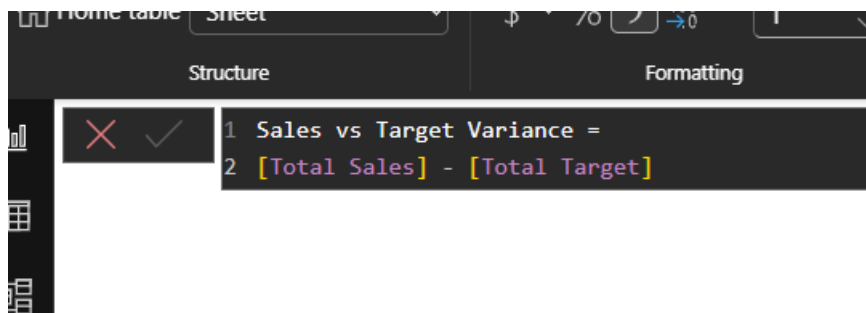
11. Create Calculated Fields for Comparison:

- **Develop new measures that leverage both datasets.**

- **Examples: Target Achievement % = (SUM(Sales) / SUM(Sales Target)) * 100**

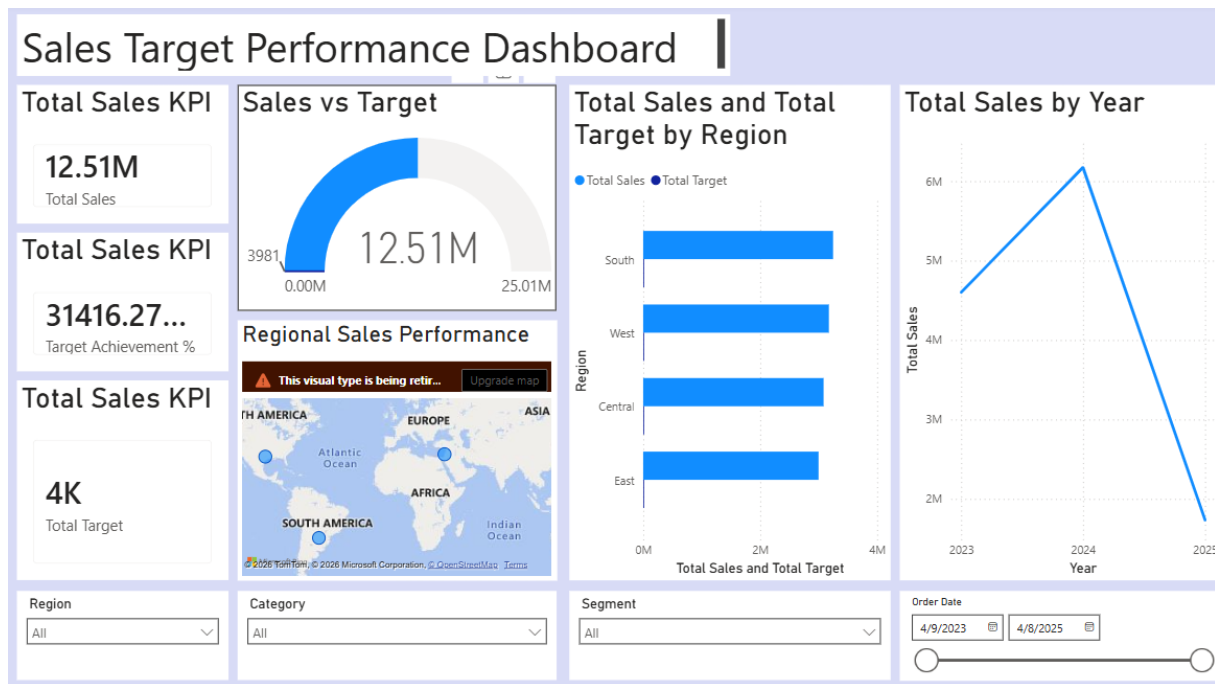


- **Sales vs. Target Variance = SUM(Sales) - SUM(Sales Target)**



12. Design a Comprehensive Dashboard:

- **Create visuals that combine data from both sources to provide a single view of performance.**
- **Add KPI cards for Total Sales and Overall Target Achievement %.**
- **Use a gauge chart or bullet chart to show sales performance against targets for each region.**
- **Create a combined bar chart showing Actual Sales vs. Target Sales by Region.**
- **Use a map to visualize the Sales vs. Target Variance, highlighting underperforming or overperforming regions.**



13. Publishing and Sharing:

- **Tableau:** Publish the workbook to Tableau Public or Tableau Server.
- **Power BI:** Publish the report to Power BI Service and share the link.

Published PowerBI

link: <https://app.powerbi.com/groups/me/reports/53422230-f178-4efb-9464-38e0e11b5e5c/d49376c1c80099bce6b1?experience=power-bi>

Experiment Datasets:

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Dataset 2: Regional Sales Targets (Excel)

Fields Used:

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- **Sales Target** – The sales goal for the region
- **Target Year** – The year for which the target is set

POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. How did integrating the sales targets data provide a more complete analytical picture than using the sales data alone?

Integrating the sales target data with the actual sales data made the analysis much more meaningful. Instead of only knowing how much sales were generated, I was able to compare the results against the planned targets. This helped in measuring overall business performance more effectively.

Some key advantages were:

- It showed whether each region met its sales target.
- It helped identify regions that were overperforming or underperforming.
- It allowed the calculation of target achievement percentage and sales variance.
- It provided better insights for monitoring business goals rather than just viewing sales figures.

Overall, combining both datasets created a more complete and informative dashboard for decision-making.

2. Describe the type of join or relationship you used to combine the two datasets and explain why it was the appropriate choice.

In Power BI, I created a Many-to-One (*:1) relationship between the Sales dataset and the Regional Sales Targets dataset using the Region field as the common key. The Sales table contained many records for each region, while the Targets table contained only one target value for each region.

This relationship was appropriate because:

- Multiple sales transactions belong to a single region.
- Each region has one corresponding sales target.
- It prevents unnecessary duplication of target values.
- It allows calculations using fields from both datasets accurately.

Using this relationship ensured that the dashboard visuals displayed correct and consistent results.

3. What challenges, if any, did you face regarding data granularity or mismatched keys (e.g., different region names) between the two sources?

One challenge I encountered was that the sales target dataset initially contained duplicate region names, which prevented Power BI from creating a proper Many-to-One relationship. I had to clean the Excel file by removing duplicate region entries before creating the relationship.

Other minor challenges included:

- Verifying that the region names matched exactly in both datasets.
- Ensuring the data types of the Region column were the same.
- Checking for missing values before creating calculations.

Cleaning and validating the data helped avoid relationship errors and ensured accurate analysis.

4. How can creating variance calculations (e.g., Sales vs. Target) help a business manager make better decisions?

Variance calculations compare the actual sales with the expected sales targets. This makes it easier for managers to understand whether business goals are being achieved and where improvements are needed.

Some benefits include:

- Quickly identifying underperforming regions.
- Recognizing regions that exceed their targets.
- Helping managers allocate resources more effectively.
- Supporting better planning for future sales strategies.
- Making performance evaluation more objective and data-driven.

Overall, variance calculations provide valuable insights that support faster and more informed

business decisions.

5. What other data source (e.g., marketing campaign data, customer demographics) could you integrate to further enrich this analysis, and what new insights might it provide?

A useful additional data source would be customer demographic data, such as customer age, gender, income level, and location. Integrating this information with the sales data would provide a deeper understanding of customer purchasing behavior.

This could help in:

- Identifying which customer groups generate the highest sales.
- Understanding buying patterns across different demographics.
- Creating targeted marketing campaigns.
- Improving product recommendations for specific customer segments.
- Supporting better business strategies based on customer preferences.

By combining demographic information with sales and target data, the dashboard would provide richer insights and enable more effective business decision-making.

RESULT:

The retail sales and regional targets datasets were successfully connected, cleaned, and integrated in both Tableau and Power BI. By creating relationships between the sources, a unified data model was built. The resulting comprehensive dashboard visualized sales performance against set targets, enabling variance analysis and providing deeper, context-rich insights into regional performance that were not possible with a single data source.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	5	
In Lab Exercise	10	
Post Lab Exercise	5	
Viva	10	
Total	30	
Faculty Signature		